

DJ Dinnebeil

Software Engineer | AI Engineering | Full-Stack Developer

Software engineer focused on backend development, data systems, and production-ready applications.
408-493-0402 | djdinn11@gmail.com | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript, SQL, HTML5, CSS, PHP
Frameworks & Tools: FastAPI, Flask, Streamlit, React, Redux, Node.js, SQLite, PostgreSQL, AWS S3, REST APIs
AI & Search: LangGraph, LangChain, Qdrant, OpenAI API, Cohere Rerank, Embeddings, Semantic Search, RAGAS

PROJECT EXPERIENCE

Historical Research Assistant — Python, LangGraph, Qdrant | [GitHub](#)

AI-powered research assistant for historians to search, evaluate, and validate primary source documents.

- Designed and implemented a semantic document retrieval system using LangGraph, Qdrant, and Streamlit to improve search relevance, ranking accuracy, and factual consistency.
- Built RAGAS-based evaluation workflows (Faithfulness, Context Precision, Answer Relevance) to measure and enhance response accuracy.
- Optimized search ranking and reranking through Cohere’s semantic scoring, refining query understanding and alignment with user intent.

The Amatol Game — JavaScript, HTML5, CSS | [GitHub](#)

A web-based history education game about the World War I ghost town of Amatol, NJ.

- Engineered interactive game mechanics using JavaScript with dynamic DOM manipulation for smooth gameplay.
- Implemented an asynchronous asset prefetching system using Promises to load images and audio at startup, reducing in-game latency.
- Utilized Tesseract OCR to digitize scanned historical documents into searchable text to provide access to primary source materials.

Auto Core — C++, Win32 API, SQLite3 | [GitHub](#)

A utility designed to streamline task execution on Windows 11 systems.

- Developed a keyboard manager with C++23 modules and the Windows API to remap keys to user-defined commands, improving workflow efficiency and taskbar navigation speed.
- Built a local database system utilizing SQLite3 to enable end-user personalization and customization.
- Implemented secure settings and data synchronization using OAuth2-based authentication.

MILITARY EXPERIENCE

United States Coast Guard Reserves June 2007 – April 2013
Electronics Technician

- Coordinated with users of electronic equipment to perform preventative and corrective maintenance.
- Recognized with the Exceptional Merit Award for technical excellence and readiness to aid colleagues.

EDUCATION

App Academy – Certificate of Completion for Software Engineering June 2024 – December 2024
AI Makerspace – Certificate of Completion for AI Engineering May 2025 – September 2025
Stockton University – B.S. in Computer Science | M.A. in American Studies August 2015 – August 2022
Honors: Certificates of Distinction in Honors and Service; Distinguished Research Fellowship
Coursework: Data Structures & Algorithms, Operating Systems, Databases, Networking, Software Engineering